

Nmap Room Walkthrough

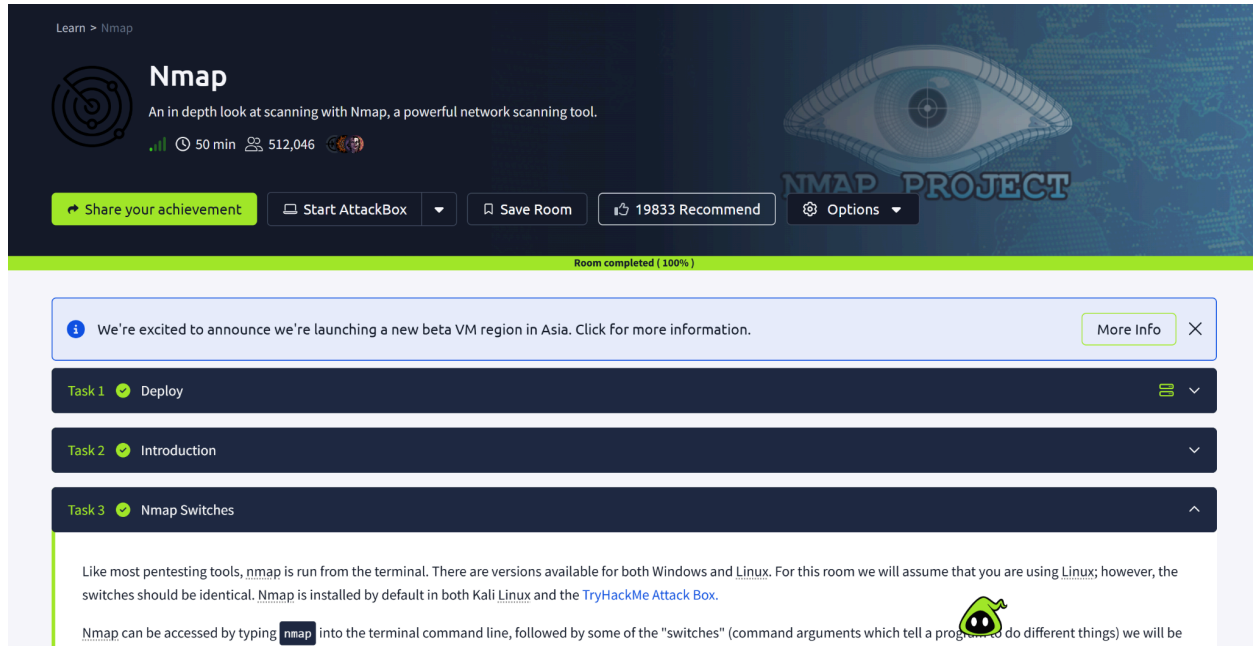
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Room URL: <https://tryhackme.com/room/furthernmap>

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Difficulty: Easy

Objective: To study and gain detailed knowledge and use in real-world scenarios to scan for vulnerabilities, and in some cases, even perform the exploit directly.



Learn > Nmap

Nmap

An in depth look at scanning with Nmap, a powerful network scanning tool.

50 min 512,046

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Task 1 Deploy

Task 2 Introduction

Task 3 Nmap Switches

Like most pentesting tools, `nmap` is run from the terminal. There are versions available for both Windows and Linux. For this room we will assume that you are using Linux; however, the switches should be identical. `Nmap` is installed by default in both Kali Linux and the TryHackMe Attack Box.

Nmap can be accessed by typing `nmap` into the terminal command line, followed by some of the "switches" (command arguments which tell a program to do different things) we will be

Introduction

In this task of TryHackMe of Nmap Room, what was provided to me was a learning path to the current industry standard tool for port scanning.

The first part introduced me to the fundamentals of Nmap,

1. Nmap Switches/Scan Types - which is a follow-up command for nmap that tells about which scan to do. (`-sS`, `-sU`, `-sV` etc)

Example: `nmap -sS -p 80`

It also provided details on

- Basic TCP port scans
- Timing templates and aggressive mode
- Advanced port scans (XMAS / FIN / NULL)
- UDP scan

It's worth noting that you would require the same "updatedb" command if you were to make your own NSE script and add it into `Nmap --` a more than manageable task with some basic knowledge of Lua!

Answer the questions below

Search for "smb" scripts in the `/usr/share/nmap/scripts/` directory using either of the demonstrated methods.
What is the filename of the script which determines the underlying OS of the SMB server?

smb-os-discovery.nse ✓ Correct Answer

Read through this script. What does it depend on?

smb-brute ✓ Correct Answer ? Hint

Task 13 ✓ Firewall Evasion

Task 14 ✓ Practical

Task 15 ✓ Conclusion

2. Nmap Scripting Engine (NSE) - written in Lua Programming language, used for scanning for vulnerabilities and automating exploitation

It was categorized into:

safe:- Won't affect the target

intrusive:- Not safe: likely to affect the target

vuln:- Scan for vulnerabilities

exploit:- Attempt to exploit a vulnerability

auth:- Attempt to bypass authentication for running services (e.g. Log into an FTP server anonymously)

brute:- Attempt to brute-force credentials for running services

discovery:- Attempt to query running services for further information about the network (e.g. query an SNMP server).

Every computer has a total of 65535 available ports; however, many of these are registered as standard ports. For example, a HTTP Webservice can nearly always be found on port 80 of the server. A HTTPS Webservice can be found on port 443. Windows **NETBIOS** can be found on port 139 and **SMB** can be found on port 445. It is important to note; however, that especially in a CTF setting, it is not unheard of for even these standard ports to be altered, making it even more imperative that we perform appropriate enumeration on the target.

If we do not know which of these ports a server has open, then we do not have a hope of successfully attacking the target; thus, it is crucial that we begin any attack with a port scan. This can be accomplished in a variety of ways – usually using a tool called **nmap**, which is the focus of this room. **Nmap** can be used to perform many different kinds of port scan – the most common of these will be introduced in upcoming tasks; however, the basic theory is this: **nmap** will connect to each port of the target in turn. Depending on how the port responds, it can be determined as being open, closed, or filtered (usually by a **firewall**). Once we know which ports are open, we can then look at enumerating which services are running on each port – either manually, or more commonly using **nmap**.

So, why **nmap**? The short answer is that it's currently the industry standard for a reason: no other port scanning tool comes close to matching its functionality (although some newcomers are now matching it for speed). It is an extremely powerful tool – made even more powerful by its scripting engine which can be used to scan for vulnerabilities, and in some cases even perform the exploit directly! Once again, this will be covered more in upcoming tasks.

For now, it is important that you understand: what port scanning is; why it is necessary; and that **nmap** is the tool of choice for any kind of initial enumeration.

Answer the questions below

What networking constructs are used to direct traffic to the right application on a server?

Ports

✓ Correct Answer

How many of these are available on any network-enabled computer?

65535

✓ Correct Answer

[Research] How many of these are considered "well-known"? (These are the "standard" numbers mentioned in the task)

1024

✓ Correct Answer

🔍 Hint



3. Firewall Evasion with NMAP

This part of the NMAP room told me about the techniques used by NMAP to evade detection and how it was used in firewall evasion.

In the room, it explained that nmap uses **-Pn**, which tells Nmap not to bother pinging the host before scanning it. This means that Nmap will always treat the target host(s) as being alive, effectively bypassing the ICMP block.

4. Practicals

In this part of Nmap it introduced me to a practical test of what I learnt throughout the room and applied it on an **ATTACK MACHINE**. The results where,

- Does the target ip respond to ICMP echo (ping) requests (Y/N)?
Ans. N
- Perform an Xmas scan on the first 999 ports of the target -- how many ports are shown to be open or filtered?
Ans. 999
- Perform a TCP SYN scan on the first 5000 ports of the target -- how many ports are shown to be open?
Ans. 5
- Open Wireshark (see [Cryillic's Wireshark Room](#) for instructions) and perform a TCP Connect scan against port 80 on the target, monitoring the results. Make

sure you understand what's going on. Deploy the `ftp-anon` script against the box. Can Nmap login successfully to the FTP server on port 21? (Y/N)
Ans. Y

Task 14 Practical

Use what you've learnt to scan the target machine and answer the following questions!

The IP address of the VM you powered on in Task1 is MACHINE_IP

(Note: If you're not a subscriber, make sure that this machine has had around ten minutes to start)

Answer the questions below

Does the target ip respond to ICMP echo (ping) requests (Y/N)?

✓ Correct Answer

Perform an Xmas scan on the first 999 ports of the target -- how many ports are shown to be open or filtered?

✓ Correct Answer

There is a reason given for this -- what is it?

Note: The answer will be in your scan results. Think carefully about which switches to use -- and read the hint before asking for help!

✓ Correct Answer

🔍 Hint

Perform a TCP SYN scan on the first 5000 ports of the target -- how many ports are shown to be open?

✓ Correct Answer

Conclusion

Throughout the TryHackMe room, I gained immense knowledge on the working and practical experience of Nmap.